



सेंद्रल ट्रान्समिशन यटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/24

Date: 04-10-2023

As per distribution list

Subject: 24th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 24th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 20th September, 2023 (Wednesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

Kashish Bhambhani
04/10/23

(Kashish Bhambhani)
General Manager (CTU)

Distribution List:

Chief Engineer (PSP&A – I) Central Electricity Authority Sewa Bhawan, R.K.Puram, New Delhi-110 066	Member Secretary Northern Regional Power Committee 18A, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi – 110 016
Director (Power System) Solar Energy Corporation of India Ltd. D-3, 1 st Floor, A wing, Religare Building, District Centre, Saket, New Delhi-110017	Director Ministry of New and Renewable Energy, Block 14, CGO Complex, Lodhi Road, New Delhi-110003
Director (SO) Grid Controller of India Limited (erstwhile Power System Operation Corporation Ltd.) 9 th Floor, IFCI Towers, 61, Nehru Place, New Delhi-110 016	Executive Director Northern Regional Load Despatch Centre 18-A, Qutab Institutional Area, Shaheed Jeet Singh Sansanwal Marg, Katwaria Sarai, New Delhi– 110 016
Director (P&C), HPPTCL, Headoffice, Himfed Bhawan,Panjari, Shimla-171005, Himachal Pradesh	Director(W&P) UP Power Transmission Company Ltd. Shakti Bhawan Extn, 3rd floor, 14, Ashok Marg, Lucknow-226 001
Director (Technical) Punjab State Transmission Corporation Ltd. Head Office, The Mall, Patiala 147001, Punjab	Director (Projects) Power Transmission Corporation of Uttarakhand Ltd. Vidyut Bhawan, Near ISBT Crossing, Saharanpur Road, Majra, Dehradun.
Development Commissioner (Power) Power Development Department Grid Substation Complex, Janipur, Jammu	Director (Technical) Rajasthan Rajya Vidyut Prasaran Nigam Ltd. Vidyut Bhawan, Jaipur, Rajasthan-302005.
Member (Power) Bhakra Beas Management Board Sector-19 B, Madhya Marg, Chandigarh - 160019	Superintending Engineer (Operation) Electricity Circle, 5 th Floor, UT Secretariat, Sector-9 D, Chandigarh - 161009
Director (Operations) Delhi Transco Ltd. Shakti Sadan, Kotla Road, New Delhi-110 002	Director (Technical) Haryana Vidyut Prasaran Nigam Ltd. Shakti Bhawan, Sector-6, Panchkula-134109, Haryana

Distribution List:

Shri Sumit Garg Senior General Manager (Commercial) AD Hydro Power Limited Bhilwara Towers, A-12, Sector-1, Noida, U.P.- 201301 Ph.: 9971096633 Email: sumitgarg@lnjbhilwara.com	
--	--

24th Consultation Meeting for Evolving Transmission Schemes in Northern Region to be held on 20.09.2023

The 24th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 20/09/2023. List of participants is enclosed at Annexure-I.

A. Application related matters in Northern Region

In the meeting, it was informed that applications received in the month of Aug'23 under GNA Regulations will be discussed in subsequent CMETS-NR meeting.

i. Transition of AD Hydro HEP under 37.6(1)

It was deliberated that transition of applicants under various categories in Northern Region were discussed in 22nd CMETS-NR held on 21.08.2023 & 23rd CMETS-NR held on 29.08.2023. In the 22nd CMETS NR under 37.6(1), it was informed that AD Hydro HEP(192 MW) had not applied for grant of GNA under 17.2 and the same shall be processed once the application is received from AD Hydro.

Subsequently, application from AD Hydro for grant of GNA of 192 MW was received on 23rd August 2023. AD Hydro HEP is presently connected at 400/220 kV Nallagarh(PG) S/s through 220 kV D/c line with one circuit LILoed at 220 kV Phozal(HP) S/s. From the studies, it is observed that, injection of power from AD Hydro at 220 kV relieves the loading of 400/220 kV ICTs at Nallagarh S/s. Therefore, it was proposed to grant GNA of 192 MW to AD Hydro with the existing system:

S. No	Name of the Applicant	Connectivity Effective (MW)	GNA to be granted through	Details of Transmission System	Start Date of GNA requested	Actual Start Date of GNA	Conn BG3 Requirement (Rs Cr.)
1	AD Hydro HEP	192	Existing System	Existing System beyond 400/220 kV Nallagarh(PG) S/s	Immediately	01.10.2023* (Refer Note-1)	Rs. 3.84 Cr. (2 lakh/MW)

Grid India informed that, sometimes due to injection of power from Phozal(HP) in peak monsoon season, the AD Hydro -Nallagarh 220 kV D/c line loading increases to 240-250 MW under outage of one circuit of AD Hydro- Nallagarh 220 kV D/c line(one circuit LILoed at Phozal(HP)) [N-1 condition]. Regarding this, CTUIL informed that as per the transition process under GNA Regulation 2022, based on the application, GNA is being granted to AD Hydro for injection at Nallagarh. However, no other GNA application from Intra state entities/plants connection at Phozal(HP) is received. Therefore treatment shall be as per CERC regulations.

Accordingly, it was agreed to grant GNA of 192 MW to M/s AD Hydro through the existing system.

Note -1 : Further, regarding the GNA start date, it was informed that, CERC vide notification dated 03.08.2023 has informed that Scheduling and Despatch of electricity with effect from 01.10.2023 shall be based on quantum of GNA, GNA^{RE}, T-GNA and T-GNA^{RE} of each of the Designated ISTS Customers (DICs) and other users of the grid in accordance with the provisions of the Central Electricity Regulatory Commission (Indian Electricity Grid Code) Regulations, 2023. Accordingly, in view of the time constraint, AD Hydro was requested to expedite the submission of requisite Conn BG-3 for the grant of GNA.

B. ISTS Expansion

i. Augmentation of 400/220 kV, 1x500 MVA (4th) ICT at 400/220kV Kankroli substation

It was informed that, at present, 400/220 kV Kankroli(PG) S/s is having transformation capacity of 3x315 MVA 400/220 kV ICTs and Connectivity was earlier granted to M/s Hindustan Zinc Limited as a Bulk Consumer during the 14th CMETS-NR meeting held on 23/12/2022.

Further during the 17th CMETS-NR meeting held on 31/03/2023, two nos. of LTA applications from M/s Serentica Renewables India 4 Private Limited, for 100 MW each, with injection point as Kallam substation (WR), for which M/s HZL was the drawee entity (as a Bulk Consumer) at Kankroli ISTS 400/220 kV substation were also discussed. In the meeting it was deliberated that based on the NRLDC inputs & CTUIL studies, considering the peak load of Kankroli complex, loading of Kankroli ICTs are observed to be N-1 non-compliant with drawl of above load (200 MW) and need of 1 No. of 500 MVA ICT (4th) ICT augmentation at Kankroli S/s for such drawl of power emerged.

In above meeting, following transmission system was agreed for drawl of power of 200 MW:

- 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s along with associated 220kV bay
- Shifting of existing 50 MVA_r Bus reactor [so as to accommodate ICT] and development of bay for reactor
- Extension of 220 kV side of ICT through 220kV Cable/GIB

Subsequently, in the 19th Consultation meeting for Evolving Transmission Schemes in NR held on 31.05.23, GNA for 200 MW (drawl of power from Kankroli (PG) S/s was agreed for grant to M/s HZL, upon completion of GNA Transition process as per the GNA Regulations. In the meeting it was deliberated that as per the present studies, 140 MW (out of 200MW load of M/s HZL) may be drawn with the existing transmission system and for the balance (60 MW) of quantum, 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s shall be required. Accordingly, M/s HZL informed that they may seek effectiveness of GNA for Part quantum in future (as per GNA regulation), based on availability of margin at that time without availability of 4th 400/220 ICT at Kankroli. However, full GNA quantum shall be made effective only with the commissioning of required 1 No. of 400/220 kV ICT at Kankroli (ISTS) S/s (GNA_{RE} schedule of M/s HZL: Jul'24). Implementation of above ICT needs to be taken after completion of GNA transition process of M/s HZL.

RVPN vide letter 27.06.23 submitted the intra state proposal to CEA/CTU for creation of 220 kV GSS Nathdwara (Upgradation) substation along with associated transmission system incl. Nathdwara – Kankroli (PG) 220kV D/c line. Regarding this, CTU informed that, with agreed grant to M/s HZL, there will be no additional margin available in Kankroli (PG) ICTs for drawl of power and ICTs may become N-1 non-compliant in peak load condition (in Kankroli complex), therefore augmentation of ICT is also required for transfer of power to Nathdwara S/s through 220kV Nathdwara -Kankroli D/c line.

Further, CTU also requested RVPN to provide firm schedule of above intra state scheme so that approval of 400/220kV Kankroli ICT may be taken up. CEA in consultation with CTU & Grid-India agreed for the proposal and same was communicated vide CEA letter to RVPN on 31.08.23. In the letter it was also informed that RVPN shall furnish the firm schedule for implementation of above proposal so that 400/220 kV, 500MVA ICT augmentation (4th) at Kankroli (PGCIL) substation may be taken up accordingly.

RVPN vide letter dated 05.09.23 informed that tender for implementation of 220kV GSS Nathdwara (upgradation) along with its associated transmission system have been floated. The techno commercial bids shall be opened on 05.09.23 and the completion period is 18 months from the date of LOA. RVPN may update the status of award.

In 23rd CMETS-NR meeting held on 29.08.23, GNA application of M/s HZL was deliberated. In the meeting it was decided that as the ICT augmentation at Kankroli(PG) shall cater to load of HZL as well as future incremental drawl requirement of RVPN from existing as well as future(Nathdwara)system, the augmentation of ICT shall be considered as common augmentation and shall be under ISTS.

In view of above deliberations, following ICT augmentation scheme was proposed in ISTS:

- 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s along with associated 220kV bay
- Shifting of existing 50 MVAr Bus reactor [so as to accommodate ICT] and development of bay for reactor
- Extension of 220 kV side of ICT through 220kV Cable/GIB

RVPN informed that for the Nathdwara transmission system tender (completion schedule is 18 months from award), Technical bids were opened on 5th Sep 2023 and the price bids are expected to be opened shortly and the. CTUIL informed that the implementation of ICT at Kankroli shall also take minimum 18 months from award. RVPN opined that since the award of Nathdwara system may take some time, it is recommended to consider Kankroli ICT for implementation after Nathdwara is awarded. RVPN informed that they will be in a position to confirm the same in first week of October'23. CTUIL asked RVPN to inform the updated status at the earliest as ICT is required both for HZL as well as Nathdwara drawl

Grid India stated that technically the proposal is in order. Accordingly, it was decided that implementation of ICT shall be taken up upon award of Nathdwara system by RVPN or submission of requisite Conn BGs by HZL for transition of 200 MW application as bulk consumer whichever is earlier.

RVPN also asked about the financial implications of the ICT augmentation at Kankroli as it shall cater to load of HZL as well as RVPN. Regarding this, CTUIL informed that treatment shall be as per the sharing regulations.

In view of above deliberations, following ICT augmentation scheme was agreed under ISTS:

- 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s along with associated 220kV bay
- Shifting of existing 50 MVAr Bus reactor [so as to accommodate ICT] and development of bay for reactor
- Extension of 220 kV side of ICT through 220kV Cable/GIB

Implementation Schedule: 18 months (to be taken up after award of Nathdwara system by RVPN or submission of requisite Conn BGs by M/s HZL for transition of 200 MW whichever is earlier)

ii. **Requirement of one set of Bus Sectionalizer at 220kV level of 400/220kV Fatehgarh-IV PS (Section-1)**

It was deliberated that transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III was discussed in 3rd meeting of NRPC(TP) held on 19.02.2021. The scheme was recommended for implementation through TCB in the 5th meeting of NCT held on 25.08.2021 and 02.09.2021. The modified Ph-III, Part A1 scheme comprised of 400/220kV 5x500MVA ICTs and 7 nos. of 220kV line bays for RE connectivity in the present scheme.

At present 2345 MW connectivity on 7 nos. of 220kV line bays were granted/agreed for grant at 220kV level of Fatehgarh-IV (Sec-1) (under implementation in Ph-III Part A1 scheme). In view of technical limitation on onward evacuation system through 400 kV Fatehgarh-IV- Fatehgarh-III (twin HLTS) D/c line(min. 2100 MVA/ckt), 2025 MW capacity (on 6 nos. of 220kV line bays) shall be evacuated through Phase-III scheme (Part A1 and other Ph-III elements) and balance 320MW capacity on 7th 220kV line bay shall be evacuated through Fatehgarh-IV (Sec-2) & associated transmission system as part of Ph-IV (Part-2 :5.5GW) scheme, which was recently approved in 14th NCT meeting. Considering above a bus sectionalizer is required at 220kV level of 400/220kV Fatehgarh-IV PS (Section-1) for reliable evacuation of RE power from Fatehgarh-IV PS (Sec-1).

CEA enquired about evacuation of power from generation project in case the generation granted connectivity at 7th bay comes up before Fatehgarh-IV PS (Sec-2) & its onward system . Regarding this, CTUIL informed that for the generation project to be granted connectivity at 7th 220kV bay, the transmission system for Connectivity under GNA shall include Fatehgarh-IV (Sec-II) & associated Phase-IV (Part-2) transmission system. Therefore, the evacuation in case of early commissioning of generation project shall be as per 22.4 of GNA Regulations through deemed T-GNA from Section-I and associated Ph-III System subject to availability of margins.

In view of above deliberations, following ISTS scheme was agreed:

- One set of bus sectionalizer at 220kV level of 400/220kV Fatehgarh-IV PS (Section-1)

Implementation Schedule: Matching with Ph-IV (Part-2: 5.5 GW scheme) (Tentative Sch. Feb'26)

iii. **Requirement of 1x80 MVar (1-Ph) spare reactor unit at Beawar S/s under Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part D scheme**

It was informed that Transmission system for Renewable Energy Zones in Rajasthan Ph-IV(Part-2 :5.5GW) scheme was approved in the 14th NCT held on 09.06.2023. Part-D of this scheme involves implementation of Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor for each circuit at each end.

765/400kV Beawar substation is being established under Phase-III Part-F scheme along associated with 330 MVar line reactor(s) & bus reactors. As there is no 80MVar spare Reactor (1-ph) available at Beawar substation in Ph-III Part-F scheme, it was proposed to consider 1x80 MVar (1-ph) spare Reactor at Beawar end under Ph-IV Part-D scheme.

Same was agreed in the meeting.

--X--

List of Participants of 24th Consultation meeting for Evolving Transmission Schemes in NR held on 20.09.2023.

CEA

Shri Nitin Deswal	Deputy Director
-------------------	-----------------

SECI

Shri R.K. Agarwal	Consultant
-------------------	------------

Grid India

Shri Gaurav Malviya	Manager
---------------------	---------

CTU

Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri Narendra Sathvik R	Manager (CTU)
Shri Roushan Kumar	Engineer (CTU)
Shri Madhusudan Meena	Engineer (CTU)

RVPN

Shri Sanjay Mathur	
Shri V. A. Kale	

HVPNL

Shri Sanjay Verma	Sr. SE (Planning)
-------------------	-------------------

HPPTCL

Shri Suresh Sharma	DGM Planning
--------------------	--------------

GNA Transition Applicants

Shri Anil Kumar Garg	AD Hydro Power Ltd.
Shri Sumit Garg	AD Hydro Power Ltd.